2.2 Inclined Plane Problems

Learning Goals (Students will be able too…)

- Solve problems with objects on inclined planes.
- Draw FBD's and break up forces into their components.

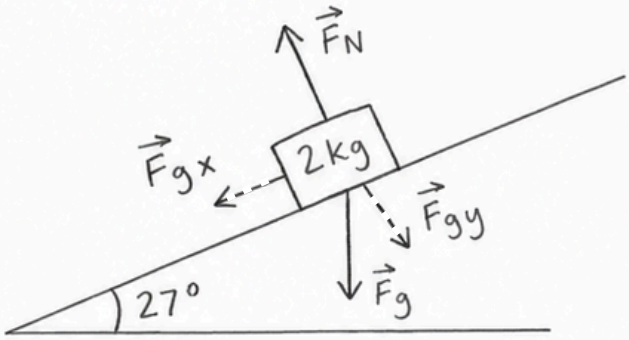
Steps for Inclined Plane Problems – Choosing our X- and Y- Directions

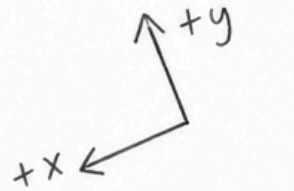
1) You always want to begin by choosing for **both the x & y component** which direction is positive and which is negative. This will make your life much easier when it comes to calculating the forces acting on both the x and the y as we say in our equilibrium lesson.
2) To simplify the analysis, always choose the positive **x direction to be in the direction of acceleration if possible.** If you do this then the acceleration in the y-direction will be __0__ (Due to Newton's Second Law)

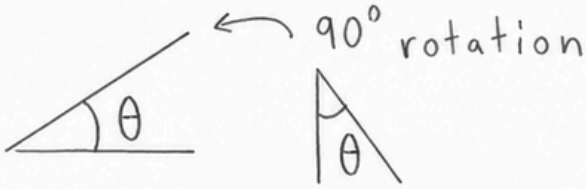
Drawing FBD with our X- and Y-Directions- Classroom Examples

Draw a FBD for the following situations, and label which direction for each component we want to be positive:

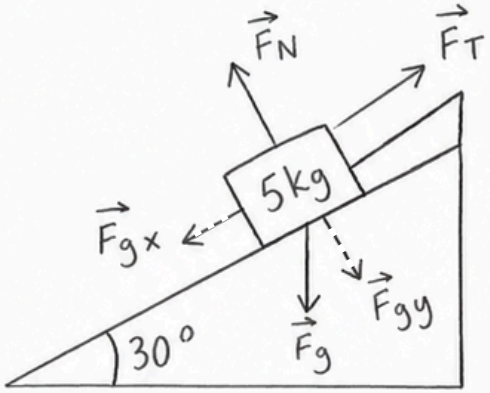
Example 1) A 2 *kg* box is put on the surface of an inclined plane at **27°** above the **horizontal** (Assume the surface of the **inclined plane** to be frictionless).

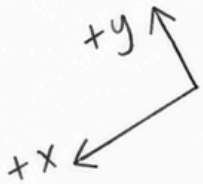






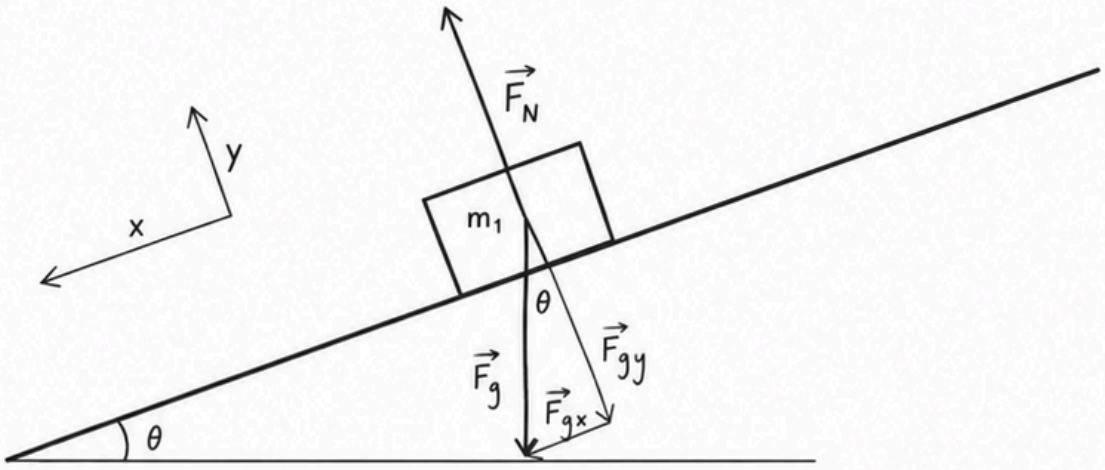
<u>Example 2)</u> An object of **5*kg*** rests on an inclined plane at **30°** above the horizontal, with a rope holding it at rest on the plane.





Typical Inclined Plane Problem

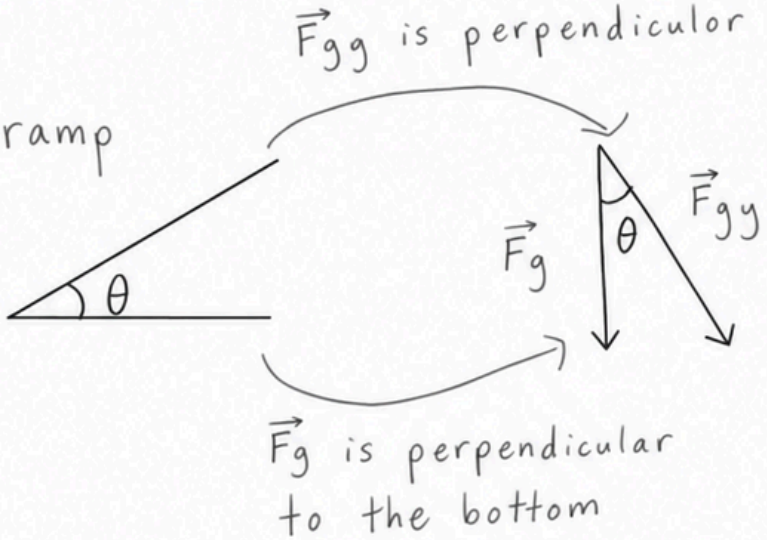
Since we are trying to always set our **positive x- direction** to be the direction of acceleration, in the case of a block sliding down a ramp, acceleration is pointing downwards, therefore we establish our coordinate system as so:



1) What force in this diagram is responsible for the objecting sliding down the ramp? (Friction Negligible).

The force of gravity, specifically the force of gravity in the x-directran (F_{gx}).

2) Using the diagram above, or re-drawing it on your paper, prove that the top angle in the force of gravity component triangle is the same as the angle of the ramp. In our last lesson, we assumed this to be true and took it for granted.



Both lines are rotated 90°, so the angle stays θ.

Inclined Plane Problems (Frictionless) – Classroom Examples

Example 1) The box on the ramp above in our 'Typical Situation' has a mass of **4.2** ***kg*** and the ramp is angled at **30°** above the horizontal.

a) Determine the component of gravity **perpendicular** to the ramp:

$F_{gy} = \cos 30^\circ (F_g)$

$F_{gy} = -36\ N$

$F_g = mg$

$F_g = 41.202\ N$

b) Why won't this component contribute to the acceleration?

Because it is perpendicular to the ramp, so the normal force opposes it and the acceleration will only be parallel to the ramp

c) Determine the component of gravity **parallel** to the ramp:

$F_{gx} = \sin 30^\circ F_g$

$F_{gx} = +21\ N$

d) Solve for acceleration.

$F_{net\,x} = F_{gx}$

$F_{net\,x} = +21\ N$

$F_{net,x} = ma_x$

$\frac{F_{net,x}}{m} = a_x$

$\frac{20.601}{4.2} = 4.905\ m/s^2$

$\vec{a} = +4.9\ m/s^2$ [down the ramp]

e) Determine the normal force.

$F_{net\,y} = F_{gy} + F_N$

$F_{net,y} = ma_y$

$F_{net\,y} = 0$

$0 = F_N + F_{gy}$

$0 = F_N - 35.68198\ N$

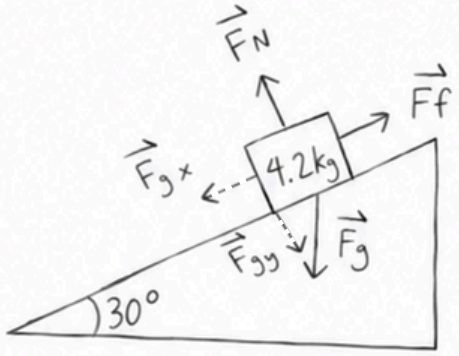
$F_N = 35.68198\ N$

$\vec{F_N} = +36\ N$

[perpendicular out of the ramp]

Example 2) Imagine the same problem as Example 1 above, but now we have friction involved.

a) What would the free body diagram of the ramp look like now?



b) The constant of kinetic friction on the ramp $\mu_k = 0.27$. (Recall that friction can be found using $F_k = \mu_k F_N$.) Determine the new acceleration of the box with friction included.

$F_{net,x} = F_{gx} + F_{fx}$ $\quad F_{net,x} = ma_x$

$F_{net,x} = +20.601 - 9.63413$ $\quad \frac{F_{net,x}}{m} = a_x$

$F_{net,x} = +10.96687\ N$ $\quad \frac{+10.96687}{4.2} = a$

$+ 2.6\ m/s^2 = \vec{a}$ [down the ramp]

$Ff = \mu_k F_N$

$Ff = 35.68198 \times 0.27$

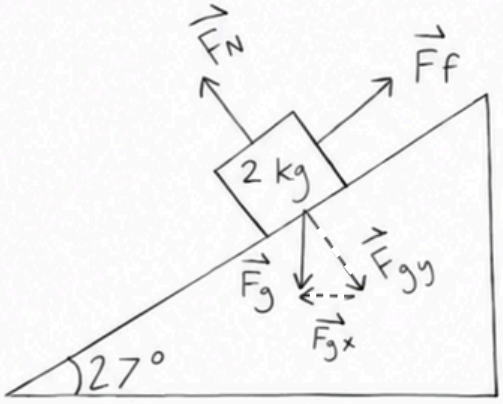
$Ff = 9.63413\ N$

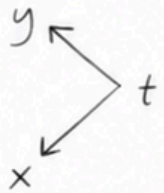
$F_{fx} = -9.6\ N$

Intro to Friction and Inclined Planes:

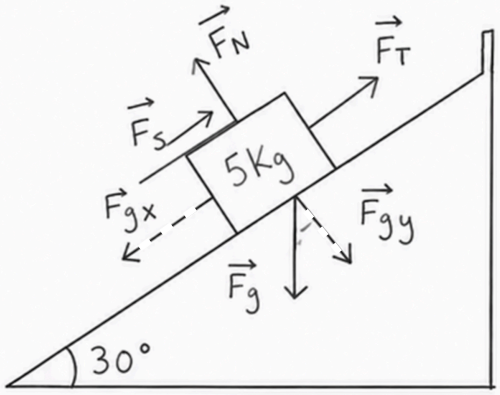
Let us re-do the FBD's for the first examples we did under "Drawing FBD with our X- and Y-Directions-Classroom Examples, but now we are considering **friction.**

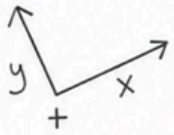
Example 1) A **2 kg** box is put on the surface of an inclined plane at **27°** above the horizontal.





Example 2) An object of $5kg$ rests on an inclined plane at **30°** above the horizontal, with a rope holding it at rest on the plane.





Discussion:

1) What component now would balance the force of gravity?
2) How does the force of static friction on the steeper ramp compare to the force of static friction of the shallower ramp? Explain.

1) $\vec{F}_N$ balances $\vec{F}_{gy}$. Along the ramp, $F_T + F_s = |F_{gx}|$.

2) For the same mass without a rope, static friction equals $mg\sin\theta$ while at rest, up to $F_{s,max} = \mu_s F_N$. A steeper ramp requires more friction.

The co-efficient of static friction can **vary up to a certain maximum value** that is determined by the material of the surfaces in contact. The **co-efficient of static friction** increases to balance gravity (x) until it can't increase any more. We are going to be exploring this concept in the next lesson where we look at static friction versus kinetic friction.

Homework

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